

an explanation linking:

the temp rises because the (rate of) energy supplied is greater than the (rate of) energy being radiated. (1)

accept 'absorbed/in' for supplied.

energy in greater than energy out

lamp radiates more as temperature increases

at constant temperature the lamp radiates energy at the same **rate** at which it is being supplied (2)

lamp radiates at the same **power** at which it is being supplied for 2 marks

power out = **power** in

award 1 mark for at constant temperature the energy supplied = energy radiated